

- 1 A curve has the equation $y^2 = kx$ for some constant $k > 0$. The point $P(a, b)$, where $a > 0$ and $b > 0$, lies on the curve. The point Q , which also lies on the curve, has an x -coordinate of $9a$ and a positive y -coordinate. What is the y -coordinate of Q in terms of b ?

- A $3b$
- B $9b$
- C $81b$
- D $\frac{b}{3}$
- E $\frac{b}{9}$
- F $3b^2$

- 2 Which one of the following is a simplification of

$$\frac{2x}{x-3} - \frac{x^2+3x}{9-x^2} - 1$$

for all values of x for which the expression is defined?

- A $2 + \frac{9}{x-3}$
- B $2 - \frac{9}{x-3}$
- C $2 + \frac{3}{x-3}$
- D $2 - \frac{3}{x-3}$
- E $1 + \frac{3}{x-3}$
- F $1 - \frac{3}{x-3}$

- 3 Find the value of x that satisfies the equation:

$$4^{\frac{1}{x-3}} \times 2^{\frac{1}{x+3}} = 16^{\frac{x-2}{x^2-9}}$$

- A $-\frac{5}{2}$
- B $\frac{5}{7}$
- C $\frac{5}{2}$
- D 4
- E 5
- F 11

- 4** A material is composed of a mixture of two types of spherical particles, type A and type B. The ratio of the number of particles of type A to the number of particles of type B is 4 : 1.

The total volume occupied by all particles of type A is half the total volume occupied by all particles of type B.

What is the ratio of the radius of a single particle of type A to the radius of a single particle of type B?

- A** $\frac{1}{8}$
- B** $\frac{\sqrt{2}}{4}$
- C** $\frac{1}{2}$
- D** $\sqrt{2}$
- E** 2
- F** 8

- 5** A point P lies on the hypotenuse AB of a right-angled triangle ABC . The perpendicular distance from P to the leg AC is equal to the perpendicular distance from P to the leg BC . Given that $AP = 3$ and $PB = 4$, what is this distance?

- A** $12/7$
- B** $12/5$
- C** 2.5
- D** $2\sqrt{3}$
- E** $7\sqrt{2}/4$

- 6** A solid cuboid has side lengths 2^a , 2^b , and 2^c , where a , b , and c are integers such that $a > b > c > 0$.

The following process is applied repeatedly to a single piece.

- Cut that piece in half by a plane perpendicular to its longest edge. If two or more edges share the longest length, choose exactly one of them.
- Keep exactly one of the two resulting pieces and discard the other.
- Continue with the kept piece.

The process stops as soon as the kept piece is a cube.

Let N be the total number of cuts required, and let S be the side length of the final cube.

Which pair of expressions for N and S is correct?

- A** $N = a - b, \quad S = 2^b$
- B** $N = a - c, \quad S = 2^c$
- C** $N = a + b - c, \quad S = 2^c$
- D** $N = a + b - 2c, \quad S = 2^c$
- E** $N = a + b - 2c, \quad S = 2^{3c}$
- 7** A tank initially contains V litres of water. Water flows into the tank at a rate of x litres per minute for a duration of h hours. Water flows out of the tank at a rate of y litres per hour for a duration of m minutes.
- Which of the following expressions gives the final volume of water in the tank, in litres?
- A** $V + hx - my$
- B** $V + 60hx - 60my$
- C** $V + \frac{hx}{60} - \frac{my}{60}$
- D** $V + 60hx - \frac{my}{60}$
- E** $V + \frac{hx}{60} - 60my$
- F** $V - 60hx + \frac{my}{60}$
- 8** The points $A(2, 6)$ and $B(6, -2)$ lie on a straight line. Point P lies on this line such that the distance AP is three times the distance PB . Given that P does not lie between A and B , what are the coordinates of P ?
- A** $(0, 10)$
- B** $(3, 4)$
- C** $(5, 0)$
- D** $(8, -6)$
- E** $(14, -18)$

- 9** A passcode of length 5 is formed by selecting distinct digits from the set $\{1, 2, 3, 4, 5, 6\}$. The digit 6 must be included in the passcode, but it cannot be the first or the last digit. How many such passcodes can be formed?
- A** 120
B 240
C 360
D 480
E 600
F 720
- 10** A particle undergoes two successive displacements. The first displacement has magnitude d and is on a bearing of 030° . The second displacement has magnitude $6\sqrt{3}$ and is on a bearing of 300° . Given that the final position of the particle is due north of its initial position, what is the value of d ?
- A** 6
B 9
C $6\sqrt{3}$
D 18
E $12\sqrt{3}$
- 11** A composite object is formed from two materials, X and Y. The density of material X is 2000 kg m^{-3} . The density of material Y is 8.0 g cm^{-3} . The volume of material X is twice the volume of material Y. What is the density of the composite object?
- A** 1600 kg m^{-3}
B 4000 kg m^{-3}
C 5000 kg m^{-3}
D 6000 kg m^{-3}

- 12 A solid cylinder has base radius r and height h . A solid cube has side length $2r$. The total surface area of the cylinder is equal to the total surface area of the cube. Which of the following expressions gives h in terms of r ?
- A $h = \left(\frac{8}{\pi} - 1\right)r$
- B $h = \frac{8r}{\pi}$
- C $h = \left(\frac{12}{\pi} - 1\right)r$
- D $h = \frac{12r}{\pi}$
- E $h = \left(\frac{12}{\pi} + 1\right)r$
- 13 Three fair, six-sided dice are rolled. Let p_S be the probability that the sum of the scores is 5. Let p_P be the probability that the product of the scores is 6. What is the value of $p_P - p_S$?
- A 0
- B $\frac{1}{216}$
- C $\frac{1}{108}$
- D $\frac{1}{72}$
- E $\frac{1}{36}$
- 14 Two regular polygons, P and Q, are such that the exterior angle of P is three times the exterior angle of Q. The sum of an interior angle of P and an interior angle of Q is 300° . What is the number of sides of polygon Q?
- A 6
- B 8
- C 12
- D 24
- E 30
- F 48

- 15** A large square with side length $\sqrt{21} + \sqrt{3}$ contains four identical shaded rectangles arranged around a central unshaded square of side length $\sqrt{21} - \sqrt{3}$, as shown in the diagram.
- What is the total area of the shaded region?

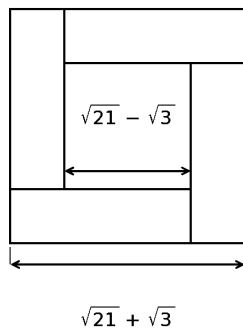


Diagram not to scale

- A** $3\sqrt{7}$
- B** $6\sqrt{7}$
- C** $12\sqrt{3}$
- D** $12\sqrt{7}$
- E** 24
- F** $24\sqrt{7}$
- G** 48

- 16** The diagram shows a circle of radius r inscribed inside a sector of a circle of radius R . The central angle of the sector is 2θ , where $0 < \theta < \frac{\pi}{2}$. The inscribed circle is tangent to both straight edges of the sector and to the circular arc. Which of the following is an expression for r in terms of R and θ ?

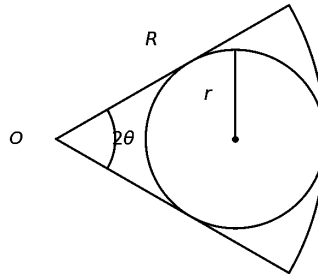


Diagram not to scale

- A** $\frac{R \sin \theta}{1 + \sin \theta}$
- B** $\frac{R \sin \theta}{1 - \sin \theta}$
- C** $\frac{R \cos \theta}{1 + \cos \theta}$
- D** $\frac{R \cos \theta}{1 - \cos \theta}$
- E** $\frac{R \tan \theta}{1 + \tan \theta}$
- F** $\frac{R}{1 + \sin \theta}$
- 17** The pressure of a gas sample is increased by 25%. The final pressure is 120 kPa. What was the initial pressure?
- A** 90 kPa
- B** 95 kPa
- C** 96 kPa
- D** 100 kPa
- E** 150 kPa

- 18** A list consists of 6 distinct numbers. A new number is added to the list, creating a new list with 7 numbers.

The mean of the new list is identical to the mean of the original list.

Which one of the following statements must be true?

- A** The range is unchanged.
- B** The median is unchanged.
- C** The range increases.
- D** The median is now equal to the mean.
- E** The median increases.
- F** The median decreases.

- 19** Given that p and q are real numbers such that

$$p + q = 1$$

and

$$8^p \times 27^q = 12$$

what is the value of p ?

- A** $\frac{1}{3}$
- B** $\frac{4}{19}$
- C** $\frac{1}{2}$
- D** $\frac{15}{19}$
- E** $\frac{2}{3}$

20 Sets of non-zero integers are classified as Type A or Type B.

- In a Type A set, 100% of the integers are even.
- In a Type B set, 50% of the integers are odd.

In any set of either type, 50% of the integers are positive. An integer's sign is independent of its parity.

Two disjoint sets, S_1 and S_2 , have equal size. An integer is chosen at random from the union $S_1 \cup S_2$.

What is the probability that the integer is odd and positive in Case 1, and what is this probability in Case 2?

- Case 1: Both S_1 and S_2 are Type B.
- Case 2: One of S_1, S_2 is Type A and the other is Type B.

- A** Case 1: $\frac{1}{4}$, Case 2: $\frac{1}{8}$
B Case 1: $\frac{1}{8}$, Case 2: $\frac{1}{4}$
C Case 1: $\frac{1}{2}$, Case 2: $\frac{1}{4}$
D Case 1: $\frac{1}{4}$, Case 2: $\frac{3}{8}$
E Case 1: $\frac{1}{8}$, Case 2: 0
F Case 1: $\frac{1}{4}$, Case 2: $\frac{1}{4}$

21 Element Q has three isotopes. The lightest isotope has a relative mass of 20. The isotope with the middle mass is 20% heavier than the lightest, and the heaviest isotope is 30% heavier than the lightest.

The relative abundance of the middle-mass isotope is equal to the sum of the relative abundances of the other two. The relative abundance of the lightest isotope is twice that of the heaviest isotope.

What is the relative atomic mass of Q?

- A** 22.0
B 22.5
C 23.0
D 23.5
E 24.0

- 22** Three distinct vertices are chosen at random from the eight vertices of a regular octagon. What is the probability that the triangle formed by these three vertices is right-angled?

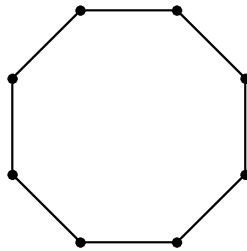


Diagram not to scale

- A** $\frac{1}{14}$
B $\frac{1}{7}$
C $\frac{2}{7}$
D $\frac{3}{7}$
E $\frac{4}{7}$
F $\frac{6}{7}$
- 23** A fluid containing a dissolved substance flows through a cylindrical pipe. Between point A and point B, the pipe's radius is halved, the fluid's speed triples, and the substance's concentration decreases by 20%. What is the percentage change in the mass of the substance flowing past a point per second, from A to B?
- A** 130% increase
B 50% increase
C 20% increase
D 25% decrease
E 40% decrease
F 60% decrease

- 24** An empty rectangular tank is being filled with water from a cylindrical pipe.
 The tank's internal dimensions are $5x$ m, $2x$ m, and y m.
 The pipe's internal diameter is $2x$ cm.
 The water flows at a constant speed of $\frac{1}{2}y$ km/h.

For positive constants x and y , find the time in hours required to fill the tank.

- A** $\frac{2}{\pi}$
B $\frac{20}{\pi}$
C $\frac{50}{\pi}$
D $\frac{200}{\pi}$
E $\frac{200\,000}{\pi}$
F $\frac{1}{50\pi}$

- 25** The graph of a quadratic polynomial $y = P(x)$ is translated by the vector $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ to give the graph of $y = Q(x)$.

The polynomial $Q(x)$ has roots at $x = 2$ and $x = 6$.

Given that the graph of $y = P(x)$ touches the x -axis exactly once, what is the y -intercept of the graph of $y = P(x)$?

- A** $-\frac{49}{2}$
B -1
C $-\frac{1}{2}$
D $\frac{1}{2}$
E 6
F $\frac{49}{2}$

- 26** The quantities x , y , P , Q and R in a physical system are related by the equation:

$$y = \frac{Px + Q}{x - R}$$

Which expression gives x in terms of y , P , Q and R ?

- A** $x = \frac{yR + Q}{P - y}$
B $x = \frac{Q + R}{y - P}$
C $x = \frac{yR - Q}{y - P}$
D $x = \frac{yR + Q}{y - P}$
E $x = \frac{yR + Q}{y + P}$

- 27** Which one of the following is a simplification of $(3\sqrt{2} - 2\sqrt{3})^2$?

- A** 6
B 30
C $6 - 12\sqrt{6}$
D $12 - 12\sqrt{6}$
E $30 - 6\sqrt{6}$
F $30 - 12\sqrt{6}$
G $30 + 12\sqrt{6}$
H $6\sqrt{2} - 4\sqrt{3}$